ITMTB-B44-Formative-EDUV4955673

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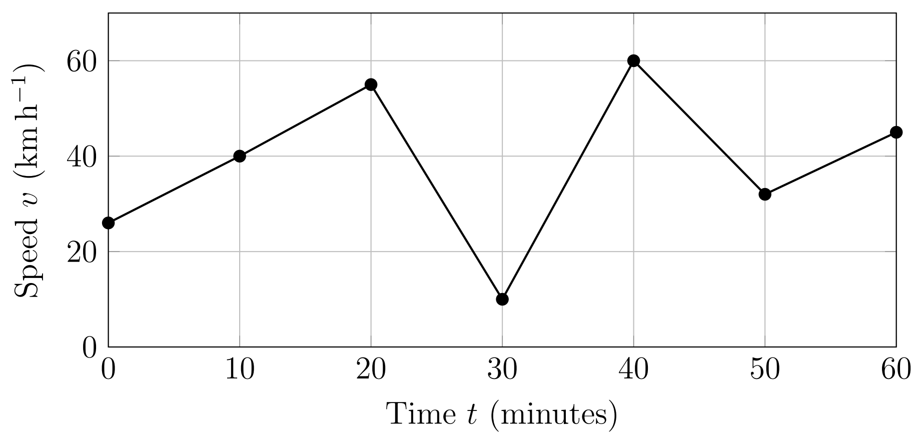
Given data

Speedometer readings (time t in minutes, speed v in $\mathrm{km} \cdot \mathrm{h}^{-1}$):

t (min)	0	10	20	30	40	50	60
v ($\mathrm{km\,h^{-1}}$)	26	40	55	10	60	32	45



1.1 (b) Approximating total distance using Riemann sums

6 subintervals of length $\Delta t = 10$ minutes $= \frac{10}{60} = \frac{1}{6}$ hour.

Left Riemann sum $L(6)$

Left endpoints: $v(0), v(10), \ldots, v(50) = 26, 40, 55, 10, 60, 32$.

$$L(6) = \sum_{i=0}^{5} v(t_i)\,\Delta t = (26 + 40 + 55 + 10 + 60 + 32) \cdot \frac{1}{6} = \frac{223}{6} \text{ km.}$$

Numerical value: $L(6) = \dfrac{223}{6} \approx 37.17$ km.

Right Riemann sum $R(6)$

Right endpoints: $v(10), v(20), \ldots, v(60) = 40, 55, 10, 60, 32, 45$.

$$R(6) = \sum_{i=1}^{6} v(t_i)\,\Delta t = (40 + 55 + 10 + 60 + 32 + 45) \cdot \frac{1}{6} = \frac{242}{6} = \frac{121}{3} \text{ km}.$$

Numerical value: $R(6) = \dfrac{121}{3} \approx 40.33$ km.

Midpoint rule $M(6)$

We use the midpoint speeds at $t = 5, 15, 25, 35, 45, 55$ minutes.

$$\begin{aligned} v(5) &= \tfrac{26+40}{2} = 33, \\ v(15) &= \tfrac{40+55}{2} = 47.5, \\ v(25) &= \tfrac{55+10}{2} = 32.5, \\ v(35) &= \tfrac{10+60}{2} = 35, \\ v(45) &= \tfrac{60+32}{2} = 46, \\ v(55) &= \tfrac{32+45}{2} = 38.5. \end{aligned}$$

Then

$$M(6) = \sum_{j=1}^{6} v\Big(t_{j-\frac{1}{2}}\Big)\,\Delta t = (33 + 47.5 + 32.5 + 35 + 46 + 38.5) \cdot \frac{1}{6} = \frac{232.5}{6}.$$

Numerical value: $M(6) = \dfrac{232.5}{6} = 38.75$ km.

1.2 Runner with acceleration $a(t) = -kt$

Acceleration: $a(t) = -kt$, initial velocity $v(0) = 192$ m/min. Integrate acceleration to get velocity:

$$v(t) = v(0) + \int_0^t a(s)\,ds = 192 + \int_0^t (-ks)\,ds = 192 - \frac{k}{2}t^2.$$

The runner comes to rest at $t = 8$ minutes, so $v(8) = 0$ gives

$$0 = 192 - \frac{k}{2} \cdot 8^2 = 192 - \frac{k}{2} \cdot 64 = 192 - 32k \implies k = 6.$$

Hence velocity function is

$$v(t) = 192 - \frac{k}{2}t^2 = 192 - 3t^2 \quad \text{(m/min)}.$$

Distance covered in the interval $0 \le t \le 8$ is

$$S = \int_0^8 v(t)\,dt = \int_0^8 (192 - 3t^2)\,dt = \Big[192t - t^3\Big]_0^8 = 192 \cdot 8 - 8^3 = 1536 - 512 = 1024 \text{ m}.$$

Answer: The runner covers $\boxed{\text{1024 meters}}$ before coming to rest.

Question 2 — Calculus Problems and Solutions

2.1

Given $k(x) = \dfrac{df}{dx}$, evaluate which constant C satisfies

$$\int_4^1 k(x)\,dx = f(4) + C.$$

By the Fundamental Theorem of Calculus,

$$\int_4^1 k(x)\,dx = f(1) - f(4).$$

Hence we require

$$f(1) - f(4) = f(4) + C \implies C = f(1) - 2f(4).$$

Therefore $C = f(1) - 2f(4)$.

2.2

Define

$$F(x) = \int_2^x \frac{1}{t^3 + t}\,dt.$$

By the Fundamental Theorem of Calculus, $F'(x) = \dfrac{1}{x^3 + x} = \dfrac{1}{x(x^2+1)}$ (for $x \neq 0$).

2.3

We are given

$$\int_8^0 f(x)\,dx = 7, \qquad \int_8^2 f(x)\,dx = 5, \qquad \int_0^2 g(x)\,dx = 9.$$

Use additivity of integrals: $\int_8^0 f = \int_8^2 f + \int_2^0 f$. Thus

$$7 = 5 + \int_2^0 f(x)\,dx \implies \int_2^0 f(x)\,dx = 2.$$

We want

$$\int_2^0 (3f(x) + g(x))\,dx = 3\int_2^0 f(x)\,dx + \int_2^0 g(x)\,dx.$$

But $\int_2^0 g(x)\,dx = -\int_0^2 g(x)\,dx = -9$. Hence

$$\int_2^0 (3f + g) = 3(2) + (-9) = 6 - 9 = -3.$$

So the value is $\boxed{-3}$.

2.4

Let $f(x) = \sqrt{x+200}$ on the interval $[-200, k]$. The average value of f on $[-200, k]$ is

$$\frac{1}{k-(-200)} \int_{-200}^{k} \sqrt{x+200}\, dx = 60.$$

Make the substitution $u = x + 200$. Then when $x = -200$, $u = 0$, and when $x = k$, $u = k + 200$. The integral becomes

$$\int_{0}^{k+200} u^{1/2}\, du = \frac{2}{3}(k+200)^{3/2}.$$

Hence the average value is

$$\frac{1}{k+200} \cdot \frac{2}{3}(k+200)^{3/2} = \frac{2}{3}(k+200)^{1/2} = 60.$$

Solve for k:

$$\sqrt{k+200} = 90 \implies k + 200 = 8100 \implies k = 7900.$$

2.5

The leak rate is given by $f'(t) = 400e^{-0.01t}$ (litres per hour). The total amount spilled in the first day (24 hours) is

$$\int_{0}^{24} 400e^{-0.01t}\, dt = 400 \left[-\frac{1}{0.01} e^{-0.01t} \right]_{0}^{24} = 400(-100)\big(e^{-0.24} - 1\big).$$

Simplifying,

$$\text{Amount} = 40000\big(1 - e^{-0.24}\big) \approx 40000(1 - 0.78663) \approx 8534.9\ extlitres.$$

Notes: Exact expression is $40000(1 - e^{-0.24})$ litres; numerical value rounded to one decimal place is ≈ 8534.9 L.

Question 3 extemdash Integration exercises

3.1 (a)

Compute

$$\int 16\sqrt{x}\, \sin\big(1 + x^{3/2}\big)\, dx.$$

Let $u = 1 + x^{3/2}$. Then $du = \frac{3}{2}x^{1/2}\, dx$, so $x^{1/2}\, dx = \dfrac{2}{3} du$. Thus

$$\int 16x^{1/2} \sin(1 + x^{3/2})\, dx = 16 \cdot \frac{2}{3} \int \sin u\, du = \frac{32}{3}(-\cos u) + C = -\frac{32}{3} \cos\big(1 + x^{3/2}\big) + C.$$

3.1 (b)

Compute the integral

$$\int Ax^3 \ln(6x)\, dx, \qquad \text{(where A is a constant; A = 125)}$$

Use integration by parts: take $U = \ln(6x)$ and $dV = Ax^3\, dx$. Then $dU = \frac{1}{x}\, dx$ and $V = A\frac{x^4}{4}$. So

$$\int Ax^3 \ln(6x)\, dx = VU - \int V\, dU \qquad\qquad = \frac{A}{4}x^4 \ln(6x) - \frac{A}{4}\int x^3\, dx$$

$$= \frac{A}{4}x^4 \ln(6x) - \frac{A}{4} \cdot \frac{x^4}{4} + C$$

$$= \frac{A}{4}x^4\Big(\ln(6x) - \frac{1}{4}\Big) + C.$$

If $A = 125$, the result is

$$\boxed{\frac{125}{4}x^4\Big(\ln(6x) - \frac{1}{4}\Big) + C}.$$

3.1 (c)

Integrate termwise the expression

$$\int \Big(1 - x - 2x^{1/2} - x^{-1/2} - 2x^{3/2} + x^{-2} - x\,x^{1/2} - x^{-1/2}\Big)\, dx.$$

First simplify the integrand. Note $x\,x^{1/2} = x^{3/2}$ and there are two occurrences of $-x^{-1/2}$. Combining like terms gives

$$1 - x - 2x^{1/2} - 2x^{-1/2} - 3x^{3/2} + x^{-2}.$$

Integrate termwise:

$$\begin{aligned}
\int 1\,dx &= x,\\
\int (-x)\,dx &= -\tfrac{x^2}{2},\\
\int (-2x^{1/2})\,dx &= -2\cdot\tfrac{2}{3}x^{3/2}\\
&= -\tfrac{4}{3}x^{3/2},\\
\int (-2x^{-1/2})\,dx &= -2\cdot 2x^{1/2}\\
&= -4x^{1/2},\\
\int (-3x^{3/2})\,dx &= -3\cdot\tfrac{2}{5}x^{5/2}\\
&= -\tfrac{6}{5}x^{5/2},\\
\int x^{-2}\,dx &= -x^{-1}.
\end{aligned}$$

Therefore the antiderivative is

$$\boxed{x-\frac{x^2}{2}-\frac{4}{3}x^{3/2}-4x^{1/2}-\frac{6}{5}x^{5/2}-x^{-1}+C}.$$

3.1 (d)

Compute

$$\int \frac{x+7}{2x^2+9x-18}\,dx.$$

Factor the denominator: $2x^2+9x-18=2(x-\frac{3}{2})(x+6)$. Use partial fractions:

$$\frac{x+7}{2x^2+9x-18}=\frac{A}{x-3/2}+\frac{B}{x+6}.$$

Clearing denominators gives

$$x+7=2A(x+6)+2B\Big(x-\tfrac{3}{2}\Big)=(2A+2B)x+(12A-3B).$$

Matching coefficients yields the system $2A+2B=1$ and $12A-3B=7$. Solving gives $A=\frac{17}{30}$ and $B=-\frac{1}{15}$. Thus

$$\int \frac{x+7}{2x^2+9x-18}\,dx=\frac{17}{30}\int\frac{1}{x-3/2}\,dx-\frac{1}{15}\int\frac{1}{x+6}\,dx \quad =\frac{17}{30}\ln\big|x-\tfrac{3}{2}\big|-\frac{1}{15}\ln|x+6|+C.$$

(Equivalently $=\dfrac{1}{30}\big(17\ln|x-3/2|-2\ln|x+6|\big)+C$.)

Answers summary:

- (a) $-\dfrac{32}{3}\cos\bigl(1+x^{3/2}\bigr)+C.$
- (b) $\dfrac{A}{4}x^4\bigl(\ln(6x)-\tfrac{1}{4}\bigr)+C$ (with $A=125$ giving $\dfrac{125}{4}x^4(\ln(6x)-\tfrac{1}{4})+C$).
- (c) $x-\dfrac{x^2}{2}-\dfrac{4}{3}x^{3/2}-4x^{1/2}-\dfrac{6}{5}x^{5/2}-x^{-1}+C.$
- (d) $\dfrac{17}{30}\ln\bigl|x-\tfrac{3}{2}\bigr|-\dfrac{1}{15}\ln|x+6|+C.$

Question 4 — Solutions

4.1

We seek the area enclosed by $y=\sin(\pi x)$, $y=2x$, and $x=0$. Solve for intersection points of $\sin(\pi x)=2x$ on $x\ge 0$. Clearly $x=0$ is a solution. Also at $x=1/2$ we have $\sin(\pi/2)=1$ and $2(1/2)=1$, so $x=1/2$ is the other intersection. On $[0,1/2]$ we have $\sin(\pi x)\ge 2x$ (e.g. at $x=1/4$), so the area is

$$A=\int_0^{1/2}\bigl(\sin(\pi x)-2x\bigr)\,dx.$$

Compute the integral:

$$\int_0^{1/2}\sin(\pi x)\,dx=-\frac{1}{\pi}\Bigl[\cos(\pi x)\Bigr]_0^{1/2}=-\frac{1}{\pi}(0-1)=\frac{1}{\pi},$$

$$\int_0^{1/2}2x\,dx=\Bigl[x^2\Bigr]_0^{1/2}=\frac{1}{4}.$$

Therefore the exact area is

$$A=\frac{1}{\pi}-\frac{1}{4}.$$

Numerically, $A\approx 0.31831-0.25=0.06831$ (approx).

$$\boxed{A=\frac{1}{\pi}-\frac{1}{4}}$$

4.2

Given

$$f(x)=x^2-3x+2,\qquad g(x)=-(x^2-3x+2)=-x^2+3x-2,\quad 0\le x\le 2.$$

(a) The exact area between the curves on $[0,2]$ is

$$A=\int_0^2\bigl(f(x)-g(x)\bigr)\,dx=\int_0^2\bigl(2x^2-6x+4\bigr)\,dx.$$

Integrate termwise:

$$\int_0^2 2x^2\,dx = 2\cdot\frac{x^3}{3}\Big|_0^2 = \frac{16}{3},\qquad \int_0^2(-6x)\,dx = -6\cdot\frac{x^2}{2}\Big|_0^2 = -12,\qquad \int_0^2 4\,dx = 4x\Big|_0^2 = 8.$$

Summing:

$$A = \frac{16}{3} - 12 + 8 = \frac{16}{3} - 4 = \frac{4}{3}.$$

Thus the exact cross-sectional area is

$$\boxed{A = \frac{4}{3}}.$$

(b) The *net difference* (pointwise difference) between the two curves is

$$f(x) - g(x) = 2(x^2 - 3x + 2) = 2x^2 - 6x + 4.$$

Answers summary:

- (4.1) Exact area $= \dfrac{1}{\pi} - \dfrac{1}{4}$.
- (4.2a) Exact area $= \dfrac{4}{3}$.
- (4.2b) Net difference function $= 2x^2 - 6x + 4$; net signed area $= \dfrac{4}{3}$.

Question 5 - Volumes of Solids of Revolution

5.1

The region bounded by

$$y = \frac{e^{3x}}{\sqrt{1+e^{6x}}},\qquad x = 0,\quad x = 1,\quad y = 0,$$

is revolved about the x-axis. Using the disk method the volume is

$$V = \pi\int_0^1 y^2\,dx = \pi\int_0^1 \frac{e^{6x}}{1+e^{6x}}\,dx.$$

Since $\dfrac{e^{6x}}{1+e^{6x}} = \frac{1}{6}\dfrac{d}{dx}\big(\ln(1+e^{6x})\big)$, $V = \dfrac{\pi}{6}\Big[\ln(1+e^{6x})\Big]_0^1 = \dfrac{\pi}{6}\big(\ln(1+e^6) - \ln 2\big)$.

$$\boxed{V = \frac{\pi}{6}\big(\ln(1+e^6) - \ln 2\big)}$$

5.2

Region: $x = \sqrt{2}\sin y,\ 0 \le y \le \pi/2,\ x = 0$.

(a) Revolving about the y-axis: use washers (integrate with respect to y). Outer radius R(y)=x(y)=$\sqrt{2}\sin y, inner radius 0. Then V = \pi\int_0^{\pi/2}[R(y)]^2\,dy = \pi\int_0^{\pi/2} 2\sin^2 y\,dy$. Using $\int_0^{\pi/2}\sin^2 y\,dy = \pi/4, we obtain V = 2\pi\cdot\frac{\pi}{4} = \frac{\pi^2}{2}$.

$$\boxed{V = \frac{\pi^2}{2}}$$

(b) Method: washer/disc (integrate w.r.t. y). **Justification:** axis is vertical and curve is x as a function of y, so radius is x(y).

(c) If revolved about the x-axis use the shell method (integrate with respect to y). For horizontal strips at height y the shell radius is y and shell height is x(y). The shell formula is

$$V = 2\pi\int_0^{\pi/2} y\,x(y)\,dy = 2\pi\int_0^{\pi/2} y\sqrt{2}\sin y\,dy.$$

5.3 - Irrigation channel (shell method about y=-5/8)

Region between

$$x_{right}(y) = \frac{y^2}{2}, \qquad x_{left}(y) = \frac{y^4}{4} - \frac{y^2}{2}, \qquad 0 \le y \le 2.$$

Using horizontal shells about y=-5/8: shell radius r(y)=y+5/8 and shell height h(y)=x$_{right} - x_{left} = y^2 - \frac{y^4}{4}. Hence V = 2\pi\int_0^2 (y+\frac{5}{8})\left(y^2 - \frac{y^4}{4}\right)dy$. Evaluating the integral gives

$$I = \int_0^2 (y+\tfrac{5}{8})\left(y^2 - \frac{y^4}{4}\right)dy = 2$$

and therefore

$$V = 2\pi I = 4\pi.$$

$$\boxed{V = 4\pi}$$

5.3 (b)

If the axis is shifted to y=-2, the radius term changes: replace (y+5/8) by (y+2) in the integrand. The height term remains y$^2 - \frac{y^4}{4}$.

5.3 (c)

Revolving about the x-axis (y=0) produces a smaller volume than revolving about y=-5/8 because every shell's radius is smaller when the axis is y=0 (radius = y) than when the axis is y=-5/8 (radius = y+5/8), while the shell heights are identical.

Answers summary (Question 5):

- 5.1: $V = \frac{\pi}{6}(\ln(1+e^6) - \ln 2)$.
- 5.2(a): $V = \frac{\pi^2}{2}$; (b) washer/disc method; (c) shell method for x-axis rotation.
- 5.3(a): $V = 4\pi$; (b) replace y+5/8 by y+2; (c) rotating about x-axis gives smaller volume.